

```
In [2]: import pandas as pd
        from sklearn.model_selection import train_test_split
        from sklearn.metrics import confusion_matrix, classification_report, accuracy_score
        from sklearn.preprocessing import LabelEncoder
        from sklearn.naive_bayes import GaussianNB
```

```
In [3]: df=pd.read_csv(r"C:\Users\Dell\Iris.csv")
        df
```

```
Out[3]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
In [4]: df.isnull().sum()
```

```
Out[4]: Id          0
        SepalLengthCm  0
        SepalWidthCm   0
        PetalLengthCm  0
        PetalWidthCm   0
        Species       0
        dtype: int64
```

```
In [5]: label_encoder=LabelEncoder()
        df['Species']=label_encoder.fit_transform(df['Species'])
```

```
In [6]: x=df.drop('Species',axis=1)
        x
```

```
Out[6]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	1	5.1	3.5	1.4	0.2
1	2	4.9	3.0	1.4	0.2
2	3	4.7	3.2	1.3	0.2
3	4	4.6	3.1	1.5	0.2
4	5	5.0	3.6	1.4	0.2
...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3
146	147	6.3	2.5	5.0	1.9
147	148	6.5	3.0	5.2	2.0
148	149	6.2	3.4	5.4	2.3
149	150	5.9	3.0	5.1	1.8

150 rows × 5 columns

```
In [7]: y=df.Species
```

```
In [8]: y
```

```
Out[8]: 0      0
1      0
2      0
3      0
4      0
..
145    2
146    2
147    2
148    2
149    2
Name: Species, Length: 150, dtype: int64
```

```
In [10]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2)
```

```
In [12]: gaussian=GaussianNB()
gaussian.fit(x_train,y_train)
```

```
Out[12]:
```

▼ GaussianNB ⓘ ?

► Parameters

```
In [13]: x_test
```

Out[13]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
60	61	5.0	2.0	3.5	1.0
33	34	5.5	4.2	1.4	0.2
122	123	7.7	2.8	6.7	2.0
130	131	7.4	2.8	6.1	1.9
43	44	5.0	3.5	1.6	0.6
17	18	5.1	3.5	1.4	0.3
79	80	5.7	2.6	3.5	1.0
77	78	6.7	3.0	5.0	1.7
108	109	6.7	2.5	5.8	1.8
129	130	7.2	3.0	5.8	1.6
34	35	4.9	3.1	1.5	0.1
123	124	6.3	2.7	4.9	1.8
65	66	6.7	3.1	4.4	1.4
40	41	5.0	3.5	1.3	0.3
131	132	7.9	3.8	6.4	2.0
72	73	6.3	2.5	4.9	1.5
103	104	6.3	2.9	5.6	1.8
3	4	4.6	3.1	1.5	0.2
90	91	5.5	2.6	4.4	1.2
24	25	4.8	3.4	1.9	0.2
27	28	5.2	3.5	1.5	0.2
21	22	5.1	3.7	1.5	0.4
62	63	6.0	2.2	4.0	1.0
88	89	5.6	3.0	4.1	1.3
1	2	4.9	3.0	1.4	0.2
119	120	6.0	2.2	5.0	1.5
35	36	5.0	3.2	1.2	0.2
144	145	6.7	3.3	5.7	2.5
54	55	6.5	2.8	4.6	1.5
45	46	4.8	3.0	1.4	0.3

```
In [14]: y_test
```

```
Out[14]: 60      1
          33      0
          122     2
          130     2
          43      0
          17      0
          79      1
          77      1
          108     2
          129     2
          34      0
          123     2
          65      1
          40      0
          131     2
          72      1
          103     2
           3      0
          90      1
          24      0
          27      0
          21      0
          62      1
          88      1
           1      0
          119     2
          35      0
          144     2
          54      1
          45      0
          Name: Species, dtype: int64
```

```
In [15]: y_test_pred=gaussian.predict(x_test)
```

```
In [16]: y_test_pred
```

```
Out[16]: array([1, 0, 2, 2, 0, 0, 1, 1, 2, 2, 0, 2, 1, 0, 2, 1, 2, 0, 1, 0, 0, 0,
                1, 1, 0, 2, 0, 2, 1, 0])
```

```
In [17]: gaussian.predict([[123,7.7,2.8,6.7,2.0]])
```

```
C:\Users\Dell\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but GaussianNB was fitted with feature names
  warnings.warn(
```

```
Out[17]: array([2])
```

```
In [18]: matrix=confusion_matrix(y_test,y_test_pred)
```

```
In [19]: matrix
```

```
Out[19]: array([[12,  0,  0],
               [ 0,  9,  0],
               [ 0,  0,  9]])
```

```
In [20]: accuracy=accuracy_score(y_test,y_test_pred)
accuracy
```

```
Out[20]: 1.0
```

```
In [21]: error=1-accuracy
```

```
In [22]: error
```

```
Out[22]: 0.0
```

```
In [23]: print(classification_report(y_test,y_test_pred))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	12
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	9
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```
In [ ]:
```